

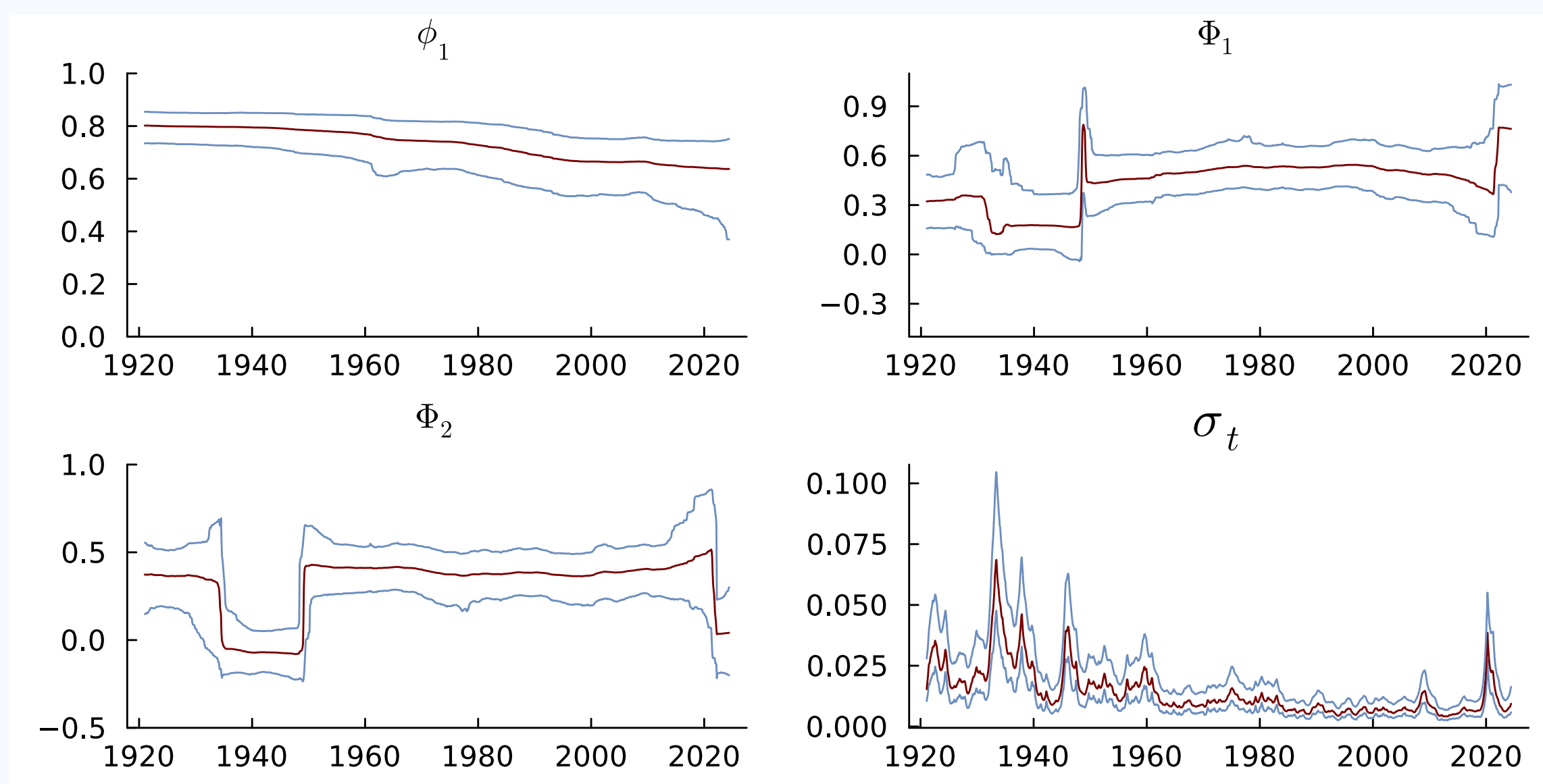
Bayesian Time-Varying Multi-Seasonal ARMA Models

Stable and Scalable Inference for Nonstationary Time Series

Ganna Fagerberg, Mattias Villani, Robert Kohn | ISBA 2026

1. Why TV-SARMA Models?

- Smooth evolution and abrupt shifts
- Multiple seasonalities (daily, weekly, annual)
- Adaptive models



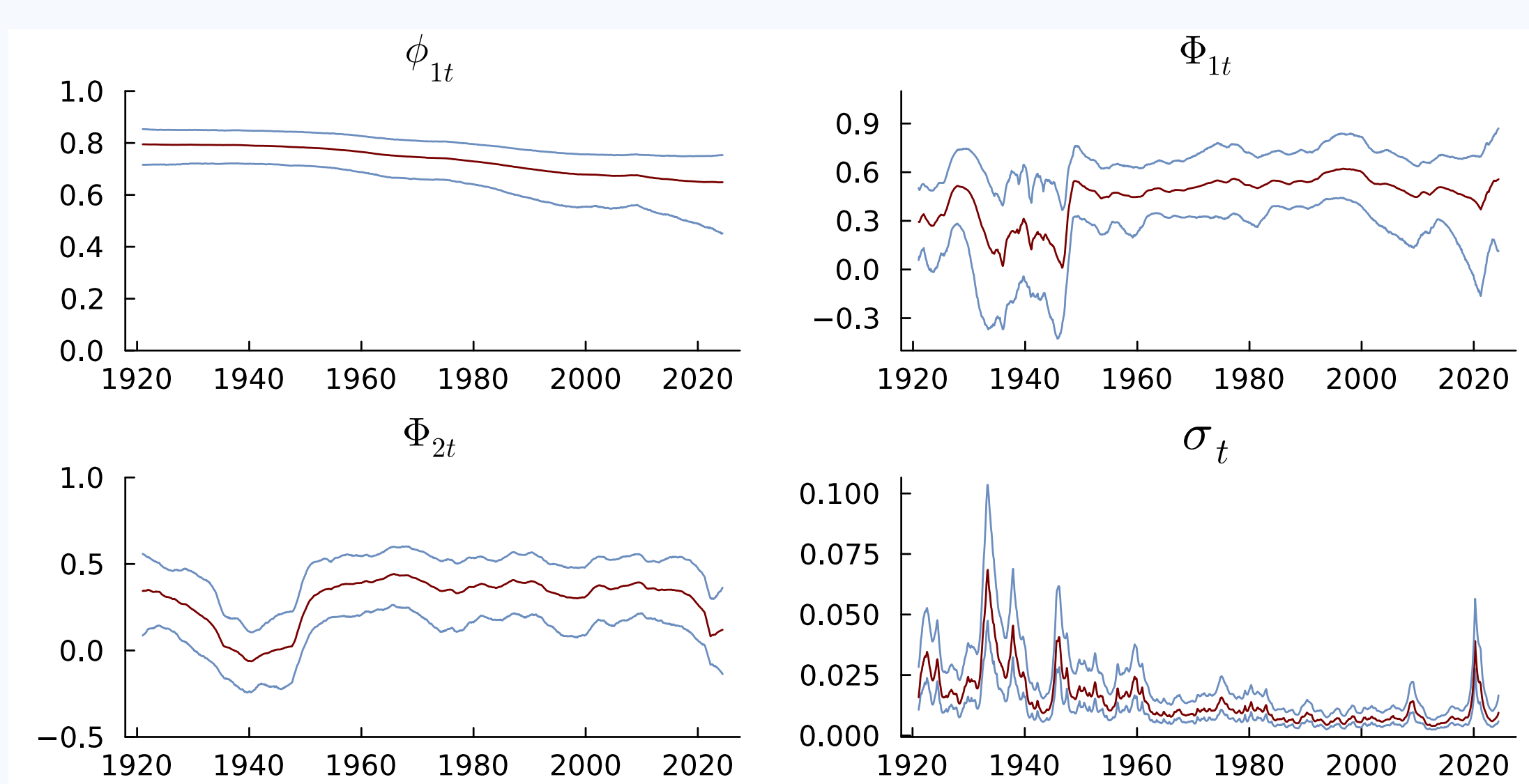
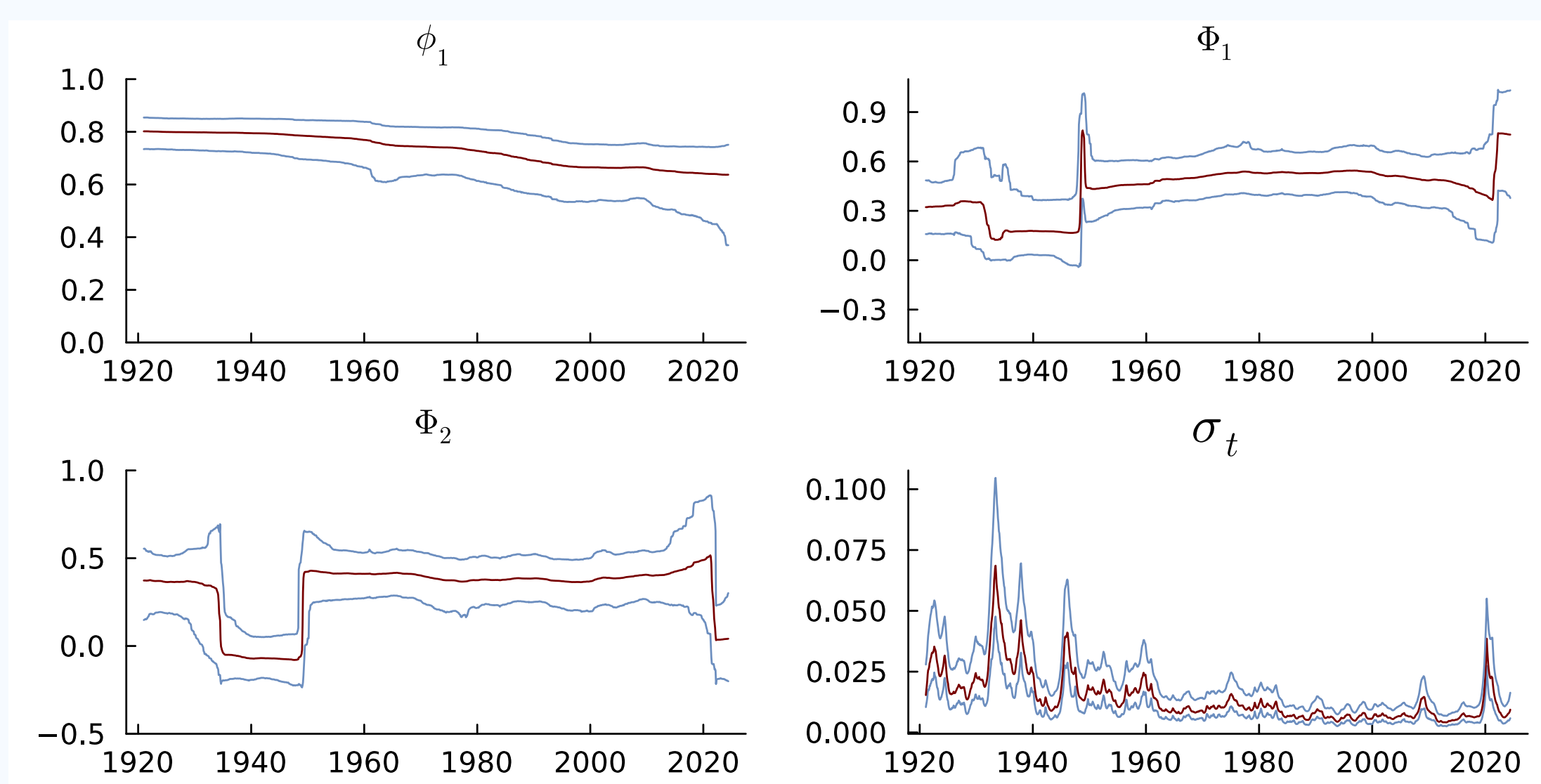
2. Modeling framework/contributions

- Bayesian TV-SARMA
- Multiple seasonalities
- Nonlinear state-space inference
- Regularization with DSP priors
- Stable and scalable inference

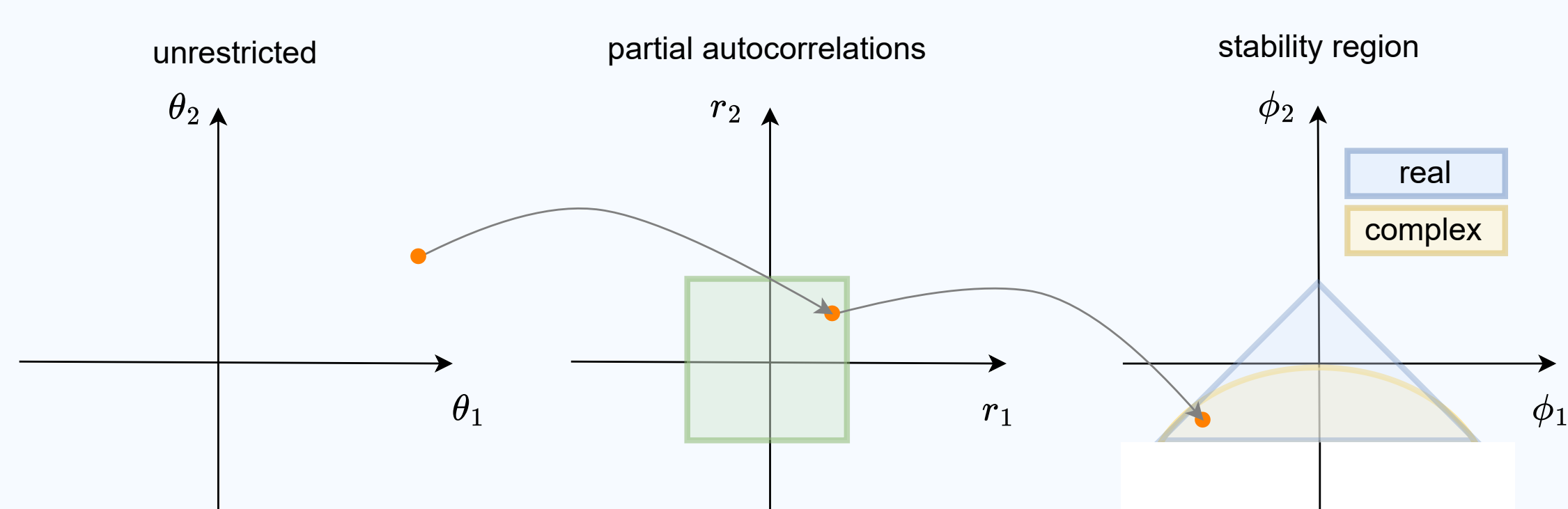
3. State-space representation

$$\begin{aligned} y_t &= \mathbf{x}_{0t}^\top \tilde{\phi}_t^{(AR)} + \mathbf{z}_{0t}^\top \tilde{\psi}_t^{(MA)} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2) \\ \tilde{\phi}_t, \tilde{\psi}_t &= \tilde{\mathbf{g}}(\theta_t) \quad \leftarrow \text{stability/invertibility constraint} \\ \theta_t &= \theta_{t-1} + \nu_t, \quad \nu_t \sim N(0, \text{Diag}(\exp(\mathbf{h}_t))) \quad \leftarrow \text{DSP prior} \\ \mathbf{h}_t &= \mu + \kappa(\mathbf{h}_{t-1} - \mu) + \eta_t, \quad \eta_t \sim Z(1/2, 1/2, 0, 1). \end{aligned}$$

4. DSP vs. Gaussian prior



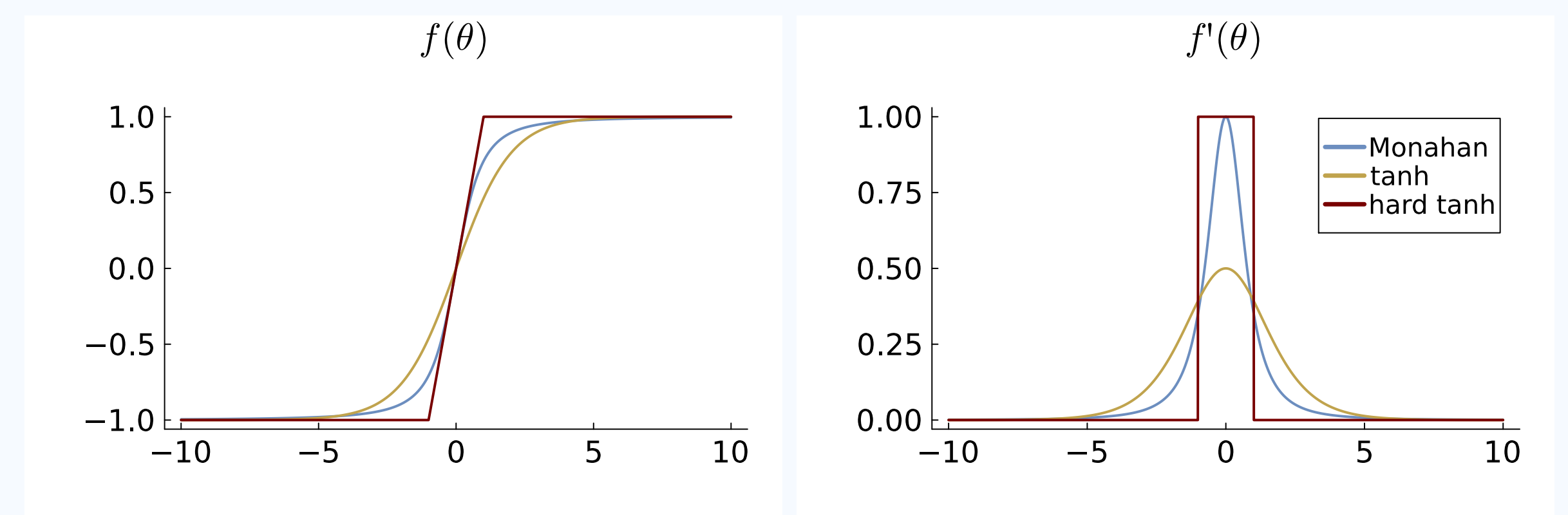
5. Stability parametrization and uniform prior



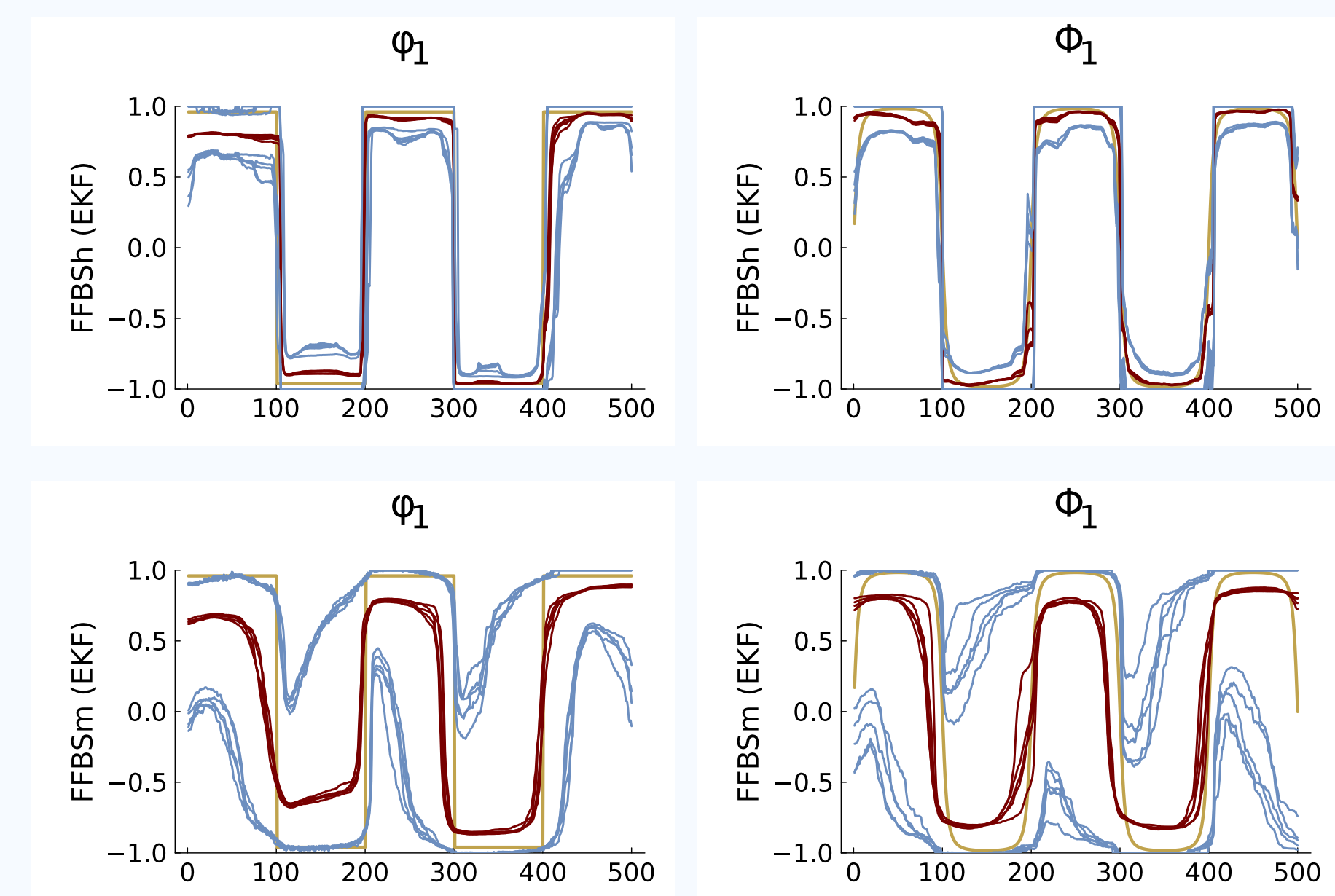
(Same idea applies to ensure invertibility!)

6. Stability parametrization at the boundary

Monahan (1984) $\Rightarrow$ tanh $\Rightarrow$ **hardtanh**



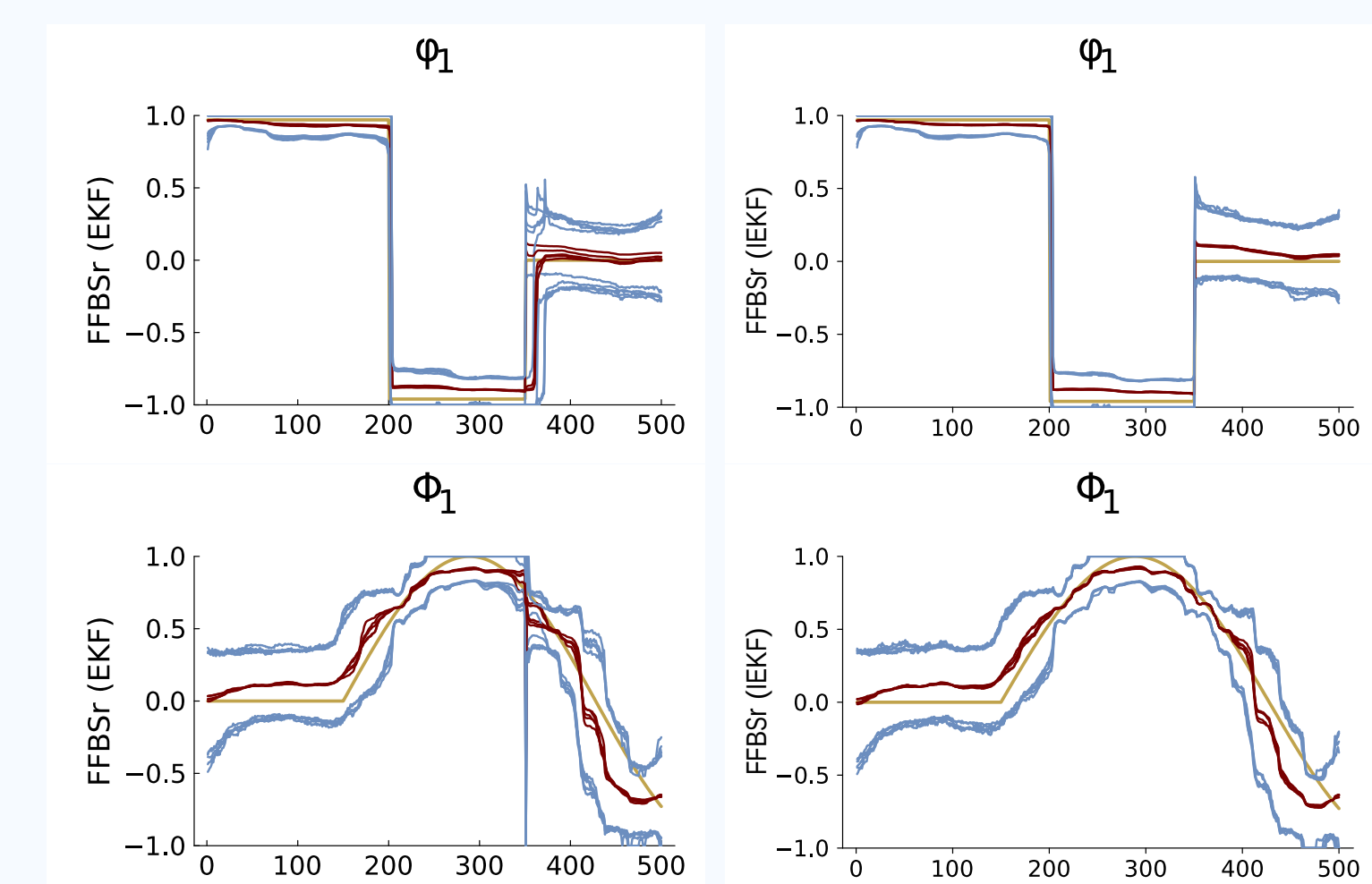
7. Parametrization effect on inference



8. Adaptive inference

Fast when possible, accurate when necessary!

FFBS+EKF $\xrightarrow{\text{NIS trigger}}$ **FFBS+IEKF**



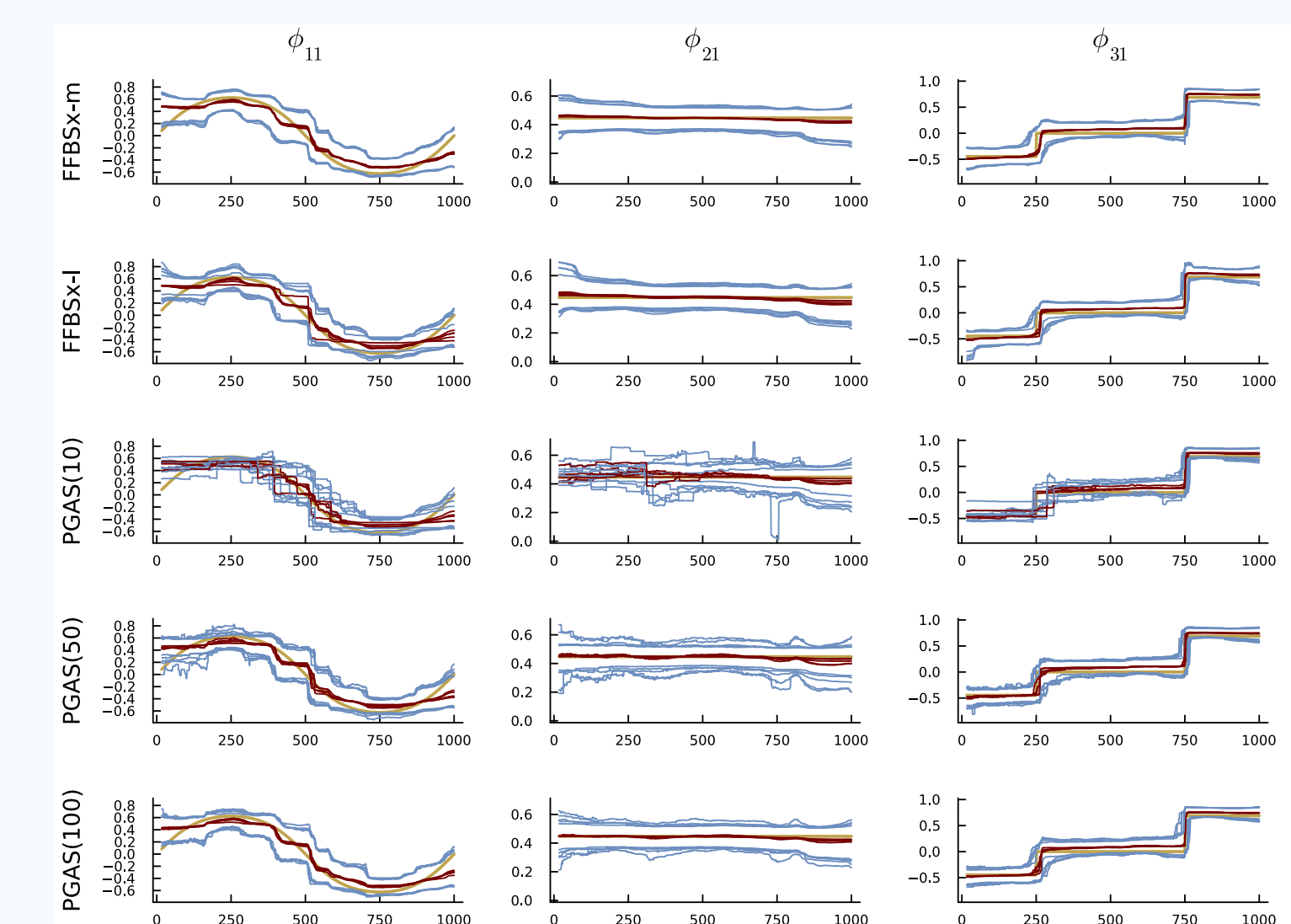
9. Gibbs sampling scheme

States $\rightarrow$ DSP $\rightarrow$ ARMA $\rightarrow$ Presample $\rightarrow$ Errors $\rightarrow$ Volatility

10. Scaling to multiple seasonalities

Grouping $\Rightarrow$ Sequential EKF $\Rightarrow$ **Scalable Inference**
(Fewer state updates) (Lower filtering cost) (Long high-frequency series)

11. Simulations



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